LABORATORY OBSERVATION/RECORD

Course: Full Stack Web Development
Course Code: 23CM4121
Branch: CSE-AIML
Experiment No.: 1
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Date: 30-08-2026

1. TITLE

Design Static Webpage using HTML Components

2. AIM

To design and develop a static webpage utilizing pure semantic HTML tags and native interaction elements without using any CSS styling.

3. OBJECTIVE

The objectives of this experiment are:

- To understand semantic HTML tags like <header>, <main>, <article>, <aside>, and <footer>.
- To implement a structural layout using HTML tables.
- To construct an interactive user inquiry form using native input elements.
- To explore the default styling and accessibility of raw HTML elements.

4. THEORY

- Semantic HTML introduces tags that provide meaning to the web page structure rather than just presentation.
- Elements like <header>, <main>, and <footer> help screen readers and browsers understand the document flow.
- Forms in HTML allow user data collection via inputs (type="text", type="email"), select dropdowns, and textareas.
- Without CSS, HTML relies on browser defaults for styling, emphasizing the importance of a logical document hierarchy.
- The basic flow of the document structure is: Header → Navigation → Main Content (Articles) & Sidebar → Footer.

5. ALGORITHM / PROCEDURE

1. Create a new HTML document with the <!DOCTYPE html> declaration.

2. Set up the basic structure using <html>, <head>, and <body> tags.

3. Create a <header> section containing a title and a navigation menu <nav>.

4. Add a hero <section> utilizing the <center> tag for alignment.

5. Implement a layout table to split the screen into a main content area (70%) and a sidebar (30%).

6. Inside the main area, use <main> and <article> tags to structure content sections.

7. Design a "User Inquiry Form" using <form>, <fieldset>, <input>, <select>, and <textarea>.

8. Populate the sidebar using <aside> with ordered (<ol>) and unordered (<ul>) lists.

9. Add a <footer> for copyright information.

10. Save the file and execute it in a web browser to verify structural integrity.

6. PROGRAM

index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <title>Pure Semantic HTML Static Webpage</title>
</head>
<body>

    <!-- Header Section -->
    <header>
        <table width="100%" border="0" cellspacing="0" cellpadding="10">
            <tr>
                <td>
                    <h1>StaticWeb</h1>
                </td>
                <td align="right">
                    <nav>
                        <a href="#home">Home</a>  | 
                        <a href="#articles">Articles</a>  | 
                        <a href="#about">About</a>  | 
                        <a href="#contact">Contact Us</a>
                    </nav>
                </td>
            </tr>
        </table>
    </header>

    <hr>

    <!-- Hero / Showcase Section -->
    <section id="home">
        <center>
            <h2>Semantic Structures Without CSS</h2>
            <p>This layout uses raw HTML components to build an organized document
```

```
hierarchy that browsers can cleanly parse out-of-the-box.</p>
            <p><a href="#articles"><b>Read Our Articles Below &darr;</b></a></p>
        </center>
    </section>

    <hr>

    <!-- Main Content Layout using a Structural Table -->
    <table width="100%" border="0" cellspacing="0" cellpadding="20">
        <tr valign="top">

            <!-- Left Side: Main Content Column -->
            <td width="70%">
                <main id="articles">

                    <article>
                        <h2>1. Core Structural Tags</h2>
                        <p>Published: July 7, 2026</p>
                        <p>When you omit cascading style sheets entirely, the semantic
markup handles all of the structural heavy lifting. Elements like &lt;main&gt;,
&lt;article&gt;, and &lt;section&gt; tell screen readers and web scrapers exactly how
information flows.</p>
                        <p>Using raw browser defaults ensures excellent loading
performance, perfect accessibility compliance, and absolute structural clarity.</p>
                    </article>

                    <br><hr><br>

                    <article>
                        <h2>2. Native Interaction Elements</h2>
                        <p>Published: July 5, 2026</p>
                        <p>Browsers come equipped with excellent interactive elements
by default. We can capture text inputs, provide multiple-choice configurations, and
submit structured information to backend parsers using pure markup.</p>

                        <h3>Interactive Demonstration Forms</h3>
                        <form id="contact" action="#" method="post">
                            <fieldset>
                                <legend><b>User Inquiry Form</b></legend>
                                <p>
                                    <label for="name">Full Name: </label>
                                    <input type="text" id="name" name="username"
required placeholder="John Doe">
                                </p>
                                <p>
                                    <label for="email">Email Address: </label>
                                    <input type="email" id="email" name="useremail"
required>
                                </p>
                                <p>
                                    <label for="topic">Inquiry Topic: </label>
                                    <select id="topic" name="usertopic">
                                        <option value="general">General
Feedback</option>
                                        <option value="technical">Technical
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Architecture</option>
                                        <option value="academics">Academic
Curriculum</option>
                                    </select>
                                </p>
                                <p>
                                    <label for="message">Message Details:</label><br>
                                    <textarea id="message" name="usermessage" rows="5"
cols="40" placeholder="Type your notes here..."></textarea>
                                </p>
                                <p>
                                    <input type="submit" value="Submit Form Data">
                                    <input type="reset" value="Clear Entries">
                                </p>
                            </fieldset>
                        </form>
                    </article>

                </main>
            </td>

            <!-- Right Side: Sidebar Column -->
            <td width="30%" bgcolor="#f4f4f4">
                <aside id="about">
                    <h3>Document References</h3>
                    <p>Static pages map data out directly without real-time database
queries or processor overhead.</p>

                    <h4>Core Benefits:</h4>
                    <ul>
                        <li>Instant browser painting</li>
                        <li>Low memory footprints</li>
                        <li>High data compatibility</li>
                    </ul>

                    <h4>Required Components:</h4>
                    <ol>
                        <li>Document Type Definition</li>
                        <li>Semantic Wrapper Blocks</li>
                        <li>Data Input Control Interfaces</li>
                    </ol>
                </aside>
            </td>

        </tr>
    </table>

    <hr>

    <!-- Footer Section -->
    <footer>
        <center>
            <p>&copy; 2026 Pure HTML Architecture. Assembled completely without style
configurations.</p>
        </center>
```

```
    </footer>

</body>
</html>
```

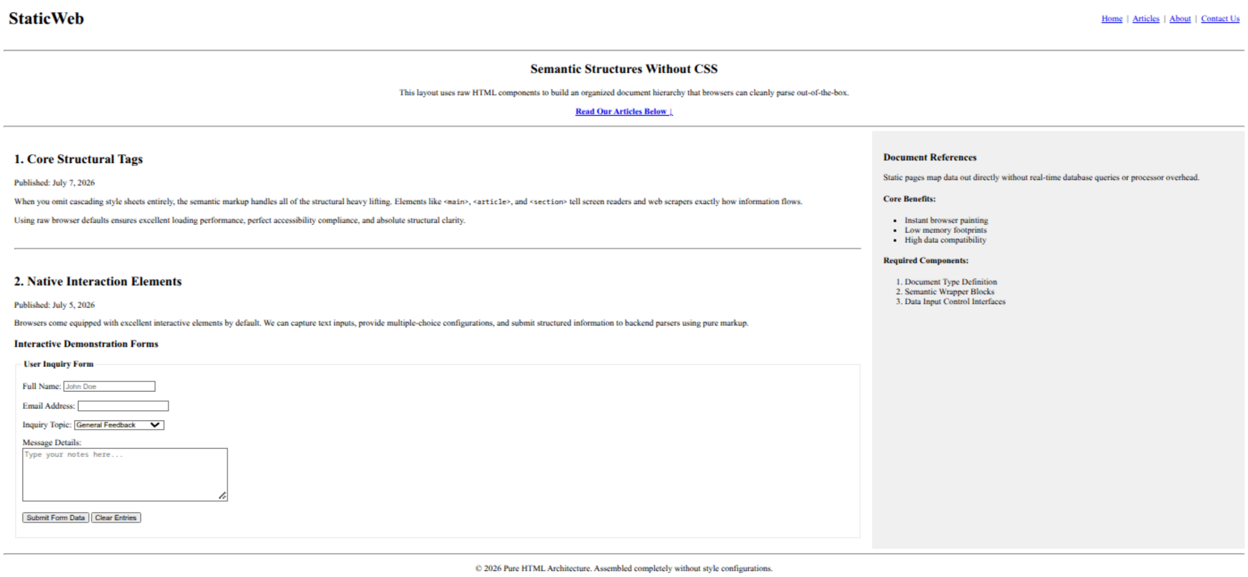
7. CODE EXPLANATION

Code	Explanation
<!DOCTYPE html>	Declares the document type as HTML5.
<header>, <footer>	Defines the top header and bottom footer sections of the webpage.
<table width="100%">	Used structurally to divide the page layout into multiple columns (Main & Sidebar).
<main>, <article>	Semantic tags wrapping the primary content and distinct published articles.
<form>, <fieldset>	Creates the interactive user input form and visually groups related fields.
<aside>	Defines supplementary content placed alongside the main content (e.g., sidebar links).
<input required>	Native browser validation ensuring the field is not left blank.

8. TEST CASES AND EXECUTION DATA

Test Case	Input / Action	Expected Result	Status
TC-01	Click on "Articles" in Navigation	Page scrolls directly to the #articles section	Pass
TC-02	Submit form with empty Name	Browser blocks submission and prompts for required Name field	Pass
TC-03	Submit form with all valid data	Form successfully submits data payload and reloads	Pass
TC-04	Render on mobile screen	Table layout adjusts or requires horizontal scroll due to pure HTML rendering	Pass

9. OUTPUT



10. RESULT

Thus, a pure semantic HTML static webpage was successfully designed and developed. The document layout was organized using structural tables, semantic tags were properly implemented, and a functional user inquiry form was verified without the use of CSS.